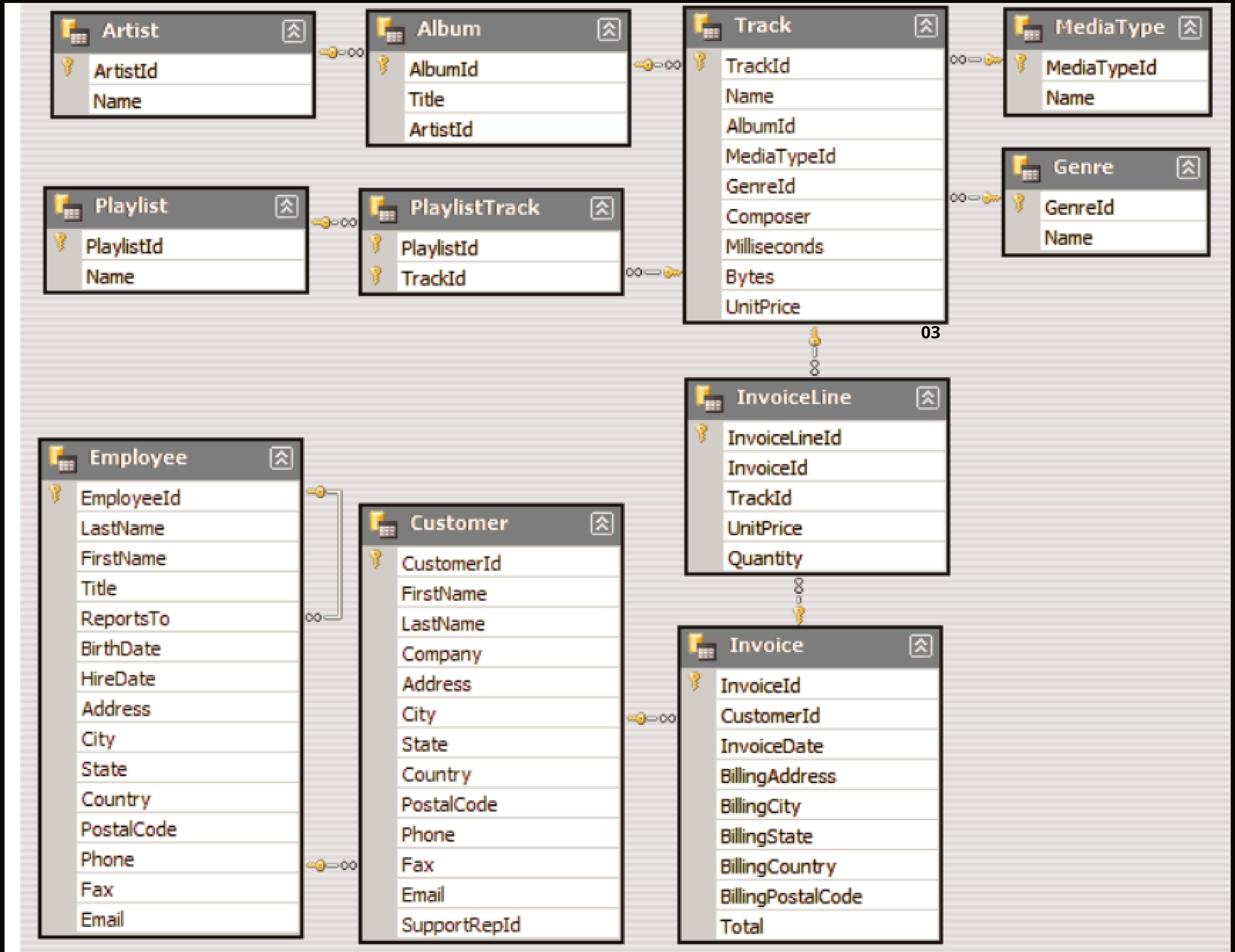
ABOUT DATASET

Analyze the MusicStoredatabase usingSQL to uncover sales trends, customer purchasing patterns, top-performing products, and revenue opportunities, demonstrating practical use of joins, aggregations and window functions for business decision support .

Database Schema Overview



Senior most employee based on the job title

```
select title ,last_name, first_name
from employee
order by levels desc
limit 1 ;
```

	title character varying (50)	last_name character (50)	first_name character (50)
1	Senior General Manag...	Madan	Mohan

Countries have the most invoices

```
select count (*)as c, billing_country
from invoice
group by billing_country
order by c desc
```

	c bigint	billing_country character varying (30)
1	131	USA
2	76	Canada
3	61	Brazil
4	50	France
5	41	Germany
6	30	Czech Republic
7	29	Portugal
8	28	United Kingdom
9	21	India

Top 3 values of total invoice

```
select total from invoice
order by total desc
limit 3
```

01

	total double precision
1	23.759999999999998
2	19.8
3	19.8

03

04

Highest sum of invoice total

```
select sum(total)as invoice_total,billing_city
from invoice
group by billing_city
order by invoice_total desc
```

	invoice_total double precision	billing_city character varying (30)
1	273.24000000000007	Prague
2	169.29	Mountain View
3	166.32	London
4	158.4	Berlin
5	151.47	Paris
6	129.69	São Paulo
7	114.83999999999997	Dublin
8	111.86999999999999	Delhi
9	108.89999999999998	São José dos Campos

Person who has spent the most money

```
select customer.customer_id,customer.first_name,customer.last_name,sum(invoice.total)as total
from customer
join invoice on customer.customer_id=invoice.customer_id
group by customer.customer_id
order by total desc
limit 1
```

	customer_id [PK] integer	first_name character (50)	last_name character (50)	total double precision
1	5	R	Madhav	144.54000000000002

Artist name & total track count of Top 10rock band

```
select artist.artist_id,artist.name,count (artist.artist_id)as number_of_songs
from track
join album on album.album_id= track.album_id
join artist on artist.artist_id= album.artist_id
join genre on genre.genre_id= track.genre_id
where genre.name like 'Rock'
group by artist.artist_id
order by number_of_songs desc
limit 10 ;
```

	artist_id [PK] character varying (50)	name character varying (120)	number_of_songs bigint
1	22	Led Zeppelin	114
2	150	U2	112
3	58	Deep Purple	92
4	90	Iron Maiden	81
5	118	Pearl Jam	54
6	152	Van Halen	52
7	51	Queen	45
8	142	The Rolling Stones	41
9	76	Creedence Clearwater Revi...	40

Return all the track names that have a song length longer than average song length

```sql
SELECT name, milliseconds
from track
where milliseconds>(
     select avg (milliseconds) as avg_track_length
     from track)
order by milliseconds desc;
```

	name character varying (150)	milliseconds integer
1	Occupation / Precipice	5286953
2	Through a Looking Glass	5088838
3	Greetings from Earth, Pt. 1	2960293
4	The Man With Nine Lives	2956998
5	Battlestar Galactica, Pt. 2	2956081
6	Battlestar Galactica, Pt. 1	2952702
7	Murder On the Rising Star	2935894
8	Battlestar Galactica, Pt. 3	2927802
9	Take the Celestra	2927677

Result

The analysis of the Music Store database provided valuable insights into sales performance, customer purchasing behavior, and product trends. Using SQL queries, the project identified top-selling tracks and genres, high-value customers, and the best-performing regions by revenue. Advanced SQL techniques, including joins, aggregations, subqueries and window functions, were used to transform raw transactional data into meaningful business insights that can support data-driven decision-making and improve business performance.

THANK YOU